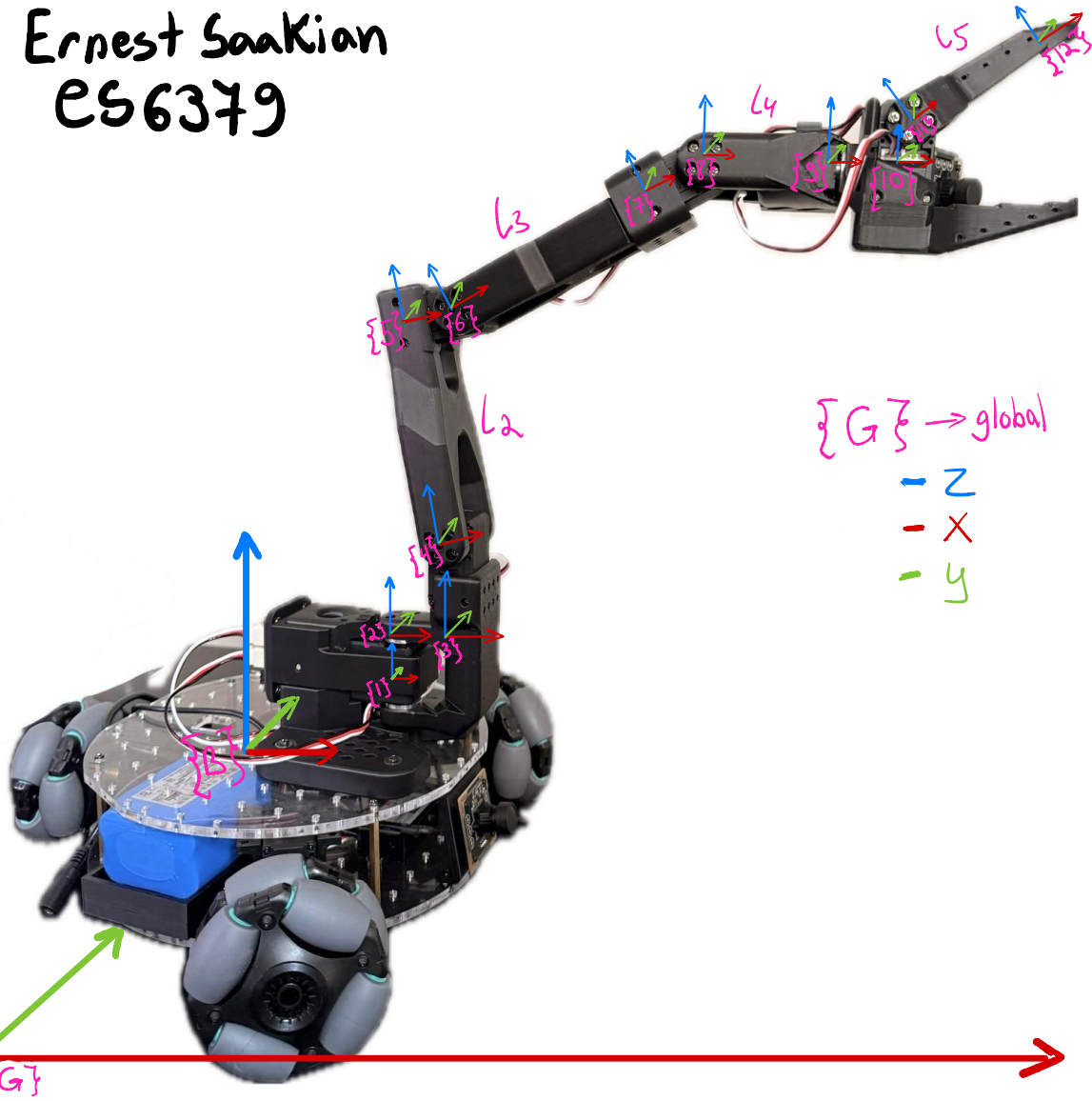


Ernest Saakian
CS 6379



Transformations

es6379 Ernest Soakian

$$T_{G12} = T_{GB} \cdot T_{B1} \cdot T_{12} \cdot T_{23} \cdot T_{34} \cdot T_{45} \cdot T_{56} \cdot T_{67} \cdot T_{78} \cdot T_{89} \cdot T_{910} \cdot T_{1011} \cdot T_{1112}$$

$$T_{GB} = \begin{bmatrix} \cos \theta_0 & -\sin \theta_0 & 0 & x \\ \sin \theta_0 & \sin \theta_0 & 0 & y \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{B1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{12} = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & t_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{23} = \begin{bmatrix} 1 & 0 & 0 & t_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{34} = \begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{45} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{56} = \begin{bmatrix} \cos \theta_3 & 0 & \sin \theta_3 & t_4 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_3 & 0 & \cos \theta_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{67} = \begin{bmatrix} 1 & 0 & 0 & l_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{78} = \begin{bmatrix} \cos \theta_4 & 0 & \sin \theta_4 & t_5 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_4 & 0 & \cos \theta_4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

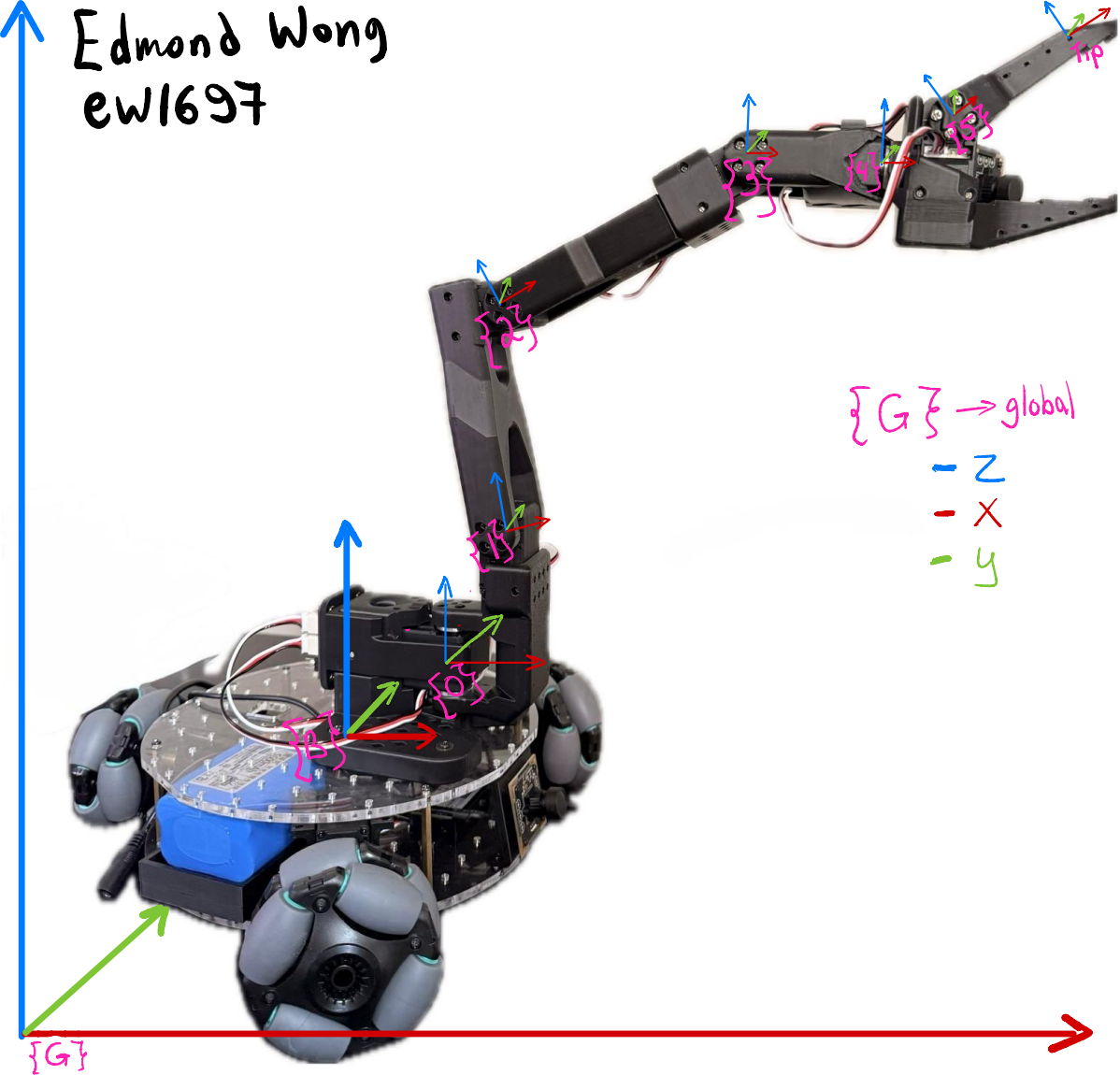
$$T_{89} = \begin{bmatrix} 1 & 0 & 0 & l_4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{910} = \begin{bmatrix} 1 & 0 & 0 & t_6 \\ 0 & \cos \theta_5 & -\sin \theta_5 & 0 \\ 0 & \sin \theta_5 & \cos \theta_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{1011} = \begin{bmatrix} \cos \theta_6 & 0 & \sin \theta_6 & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_6 & 0 & \cos \theta_6 & t_7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

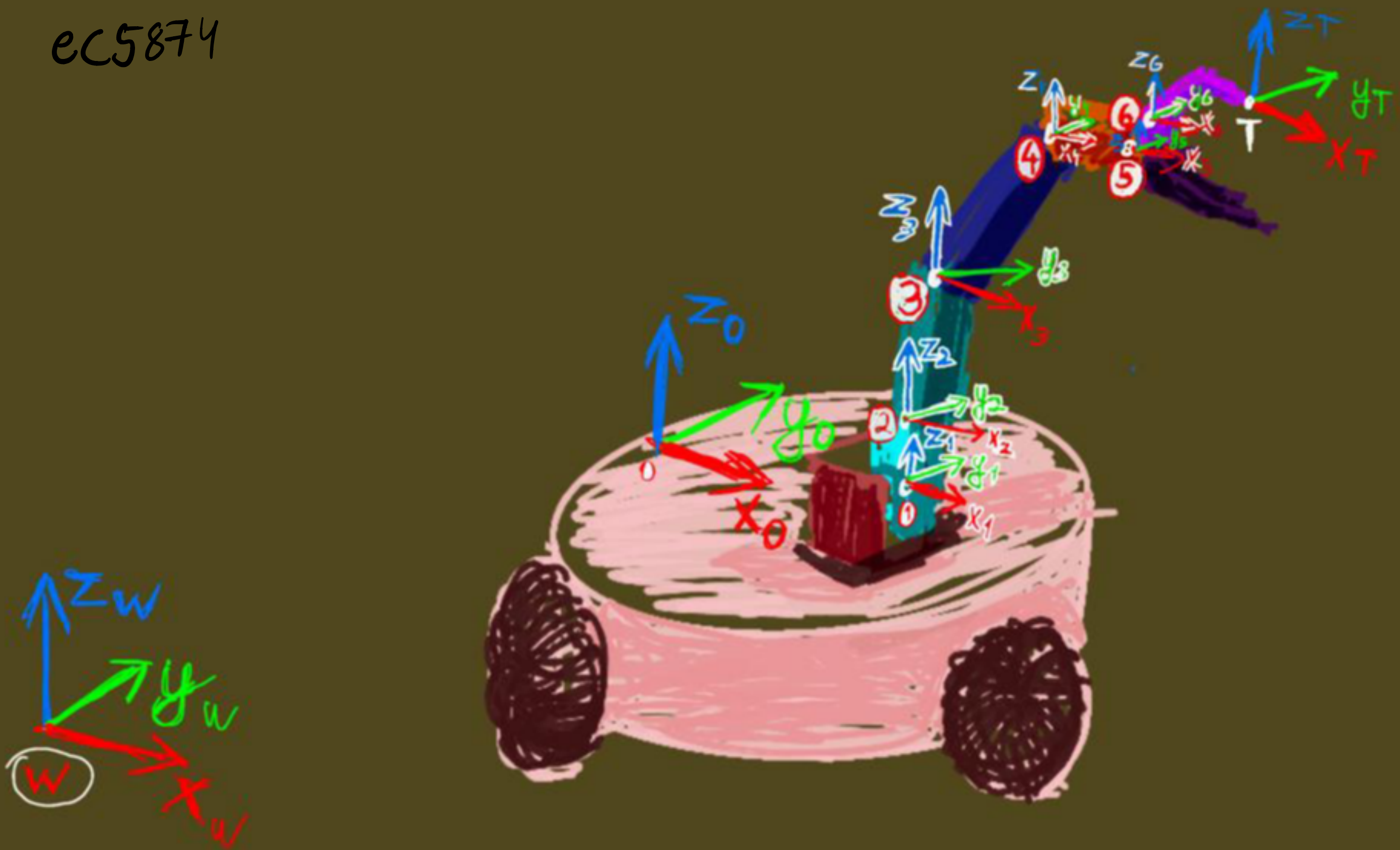
$$T_{1112} = \begin{bmatrix} 1 & 0 & 0 & l_5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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Enkel Cerri

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Suggested Title: New Note Edit
4 October 2026 at 2:10 PM

$$T = \begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix} \quad R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

rotation by z axis rotation by y axis

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

Rotation by x axis

World Frame $\rightarrow$ Robot Frame ($W \rightarrow O$)

$$P_{WO} = \begin{bmatrix} x_r \\ y_r \\ h \end{bmatrix} = \begin{bmatrix} x_r \\ y_r \\ 0,082 \end{bmatrix}$$

$$R_{WO} = R_z(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_{WO} = \begin{bmatrix} R_{WO} & P_{WO} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 & x_r \\ \sin \alpha & \cos \alpha & 0 & y_r \\ 0 & 0 & 1 & 0,082 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Joint 2 $\rightarrow$ Joint 3

$$P_{23} = \begin{bmatrix} l_3 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0,1160 \\ 0 \\ 0 \end{bmatrix}$$

$$R_{23} = R_y(\theta_3) = \begin{bmatrix} \cos \theta_3 & 0 & \sin \theta_3 \\ 0 & 1 & 0 \\ -\sin \theta_3 & 0 & \cos \theta_3 \end{bmatrix}$$

Rotation by y axis

$$T_{23} = \begin{bmatrix} \cos \theta_3 & 0 & \sin \theta_3 & 0,1160 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_3 & 0 & \cos \theta_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Joint 4 $\rightarrow$ Joint 5

$$P_{45} = \begin{bmatrix} l_5 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0,0637 \\ 0 \\ 0 \end{bmatrix}$$

$$R_{45} = R_x(\theta_5) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin \theta_5 & -\cos \theta_5 \\ 0 & \cos \theta_5 & \sin \theta_5 \end{bmatrix}$$

Rotation by x-axis

$$T_{45} = \begin{bmatrix} 1 & 0 & 0 & 0,0637 \\ 0 & \cos \theta_5 & -\sin \theta_5 & 0 \\ 0 & \sin \theta_5 & \cos \theta_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Robot FRAME $\rightarrow$ Joint 1 ($0 \rightarrow 1$)

$$P_{01} = \begin{bmatrix} 0 \\ 0 \\ l_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0,0735 \end{bmatrix}$$

$$R_{01} = R_z(\theta_1) = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation by z axis

$$T_{01} = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0,0735 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Joint 1 $\rightarrow$ Joint 2

$$P_{12} = \begin{bmatrix} 0 \\ 0 \\ l_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0,0648 \end{bmatrix}$$

$$R_{12} = R_y(\theta_2) = \begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 \\ 0 & 1 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 \end{bmatrix}$$

Rotation by y axis

$$T_{12} = \begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 & 0,0648 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Joint 3 $\rightarrow$ Joint 4

$$P_{34} = \begin{bmatrix} l_4 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0,1350 \\ 0 \\ 0 \end{bmatrix}$$

$$R_{34} = R_y(\theta_4) = \begin{bmatrix} \cos \theta_4 & 0 & \sin \theta_4 \\ 0 & 1 & 0 \\ -\sin \theta_4 & 0 & \cos \theta_4 \end{bmatrix}$$

Rotation by y-axis

$$T_{34} = \begin{bmatrix} \cos \theta_4 & 0 & \sin \theta_4 & 0,1350 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_4 & 0 & \cos \theta_4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Joint 5 $\rightarrow$ Tip

$$P_{5T} = \begin{bmatrix} l_6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0,0984 \\ 0 \\ 0 \end{bmatrix}$$

$$R_{5T} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \text{because Nothing Rolls}$$

$$T_{5T} = \begin{bmatrix} 1 & 0 & 0 & 0,0984 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{WT} = T_{WO} T_{01} T_{12} T_{23} T_{34} T_{45} T_{5T}$$